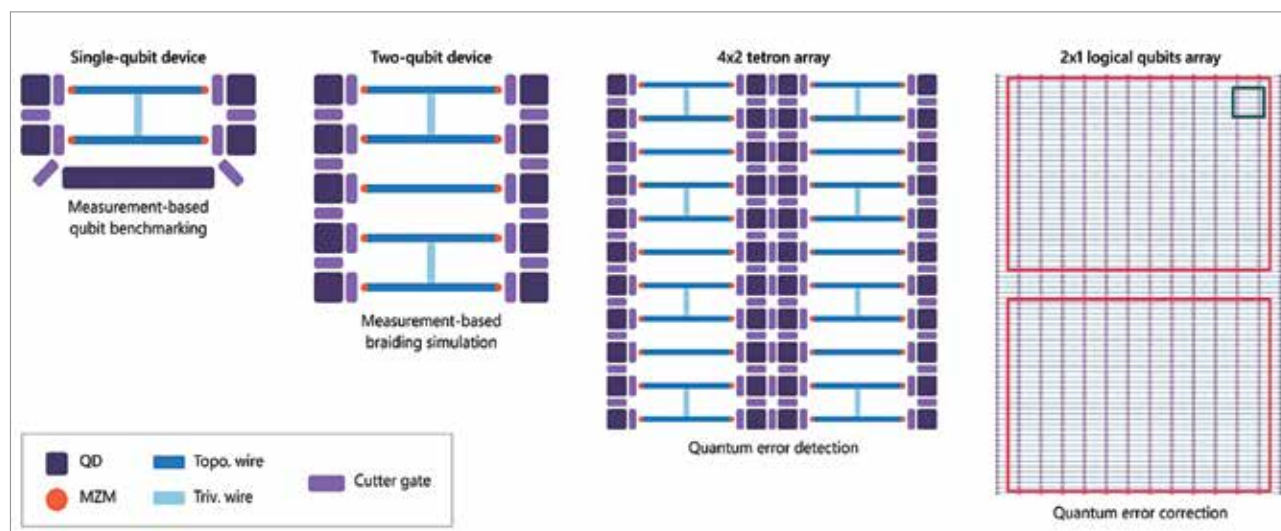




Roadmap to fault-tolerant quantum computation with tetrons. The image shows single-qubit, two-qubit, 4x2, and finally, a 27x13 tetron array (Image courtesy: Microsoft Azure)

Microsoft uses a simple measurement approach using microwaves. They reflect a microwave off a quantum dot—the way it reflects tells whether the number of electrons is even or odd. The team claims that the method is accurate enough to detect the difference between one-billion and one-billion-and-one electrons in a superconducting wire and is also extremely stable compared to traditional approaches. This simple mechanism also reduces the physical requirements, and improves the scalability of the quantum system.

#### **Scalable chip architecture:**

Microsoft's scalable architecture is built around a single-qubit device called a tetron. A tetron comprises two parallel topological wires with an MZM at each end, connected by a perpendicular trivial superconducting wire. So, basically, a tetron has four controllable Majoranas, and makes up a single qubit. Multiple tetrons can be connected and arranged across a chip. As of now, they have placed eight topological qubits on a chip, which is designed to scale to one million.

**Putting them together.** Microsoft's quantum chip fits the qubits as well as associated control electronics into a

palm-sized form factor. This, in turn, will have to be combined with control logic, a dilution refrigerator to keep the qubits at temperatures colder than outer space, and a software stack. These components are either already available or being developed in-house by Microsoft. The company hopes to build the world's first fault-tolerant prototype of a scalable quantum computer within a few years.

Just like semiconductor chips transformed the face of computing, topoconductor-based chips have the potential to build quantum systems that can scale to a million qubits. "Whatever you're doing in the quantum space needs to have a path to a million qubits. If it doesn't, you're going to hit a wall before you get to the scale at which you can solve the really important problems that motivate us," Nayak says in the blog.

#### **Amazon's quantum chip inspired by Schrodinger's cat**

A week after Microsoft announced Majorana, Amazon unveiled its prototype quantum chip, interestingly named Ocelot, after a medium-sized wild cat species! Like Microsoft, they have also built the prototype

from the ground up instead of working on improving existing error correction methods. As we saw earlier, most quantum error correction methods revolve around encoding quantum information across multiple physical qubits grouped into logical qubits to protect stored information from the environment and to detect and correct errors as they occur.

If you look at the number of qubits required to get accurate results in such methods, you will realise that it is expensive and not very scalable. Hence, the AWS quantum team decided to start from the ground up, building error correction into the hardware itself. This cuts the cost of quantum error correction by up to 90% compared to current approaches, thereby bringing down the total cost of a quantum chip significantly.

**Bosonic quantum error correction using cat qubits:** The building block of AWS' Ocelot architecture is the cat qubit, a technology inspired by the famous Schrodinger's cat thought equation. Cat qubits store quantum states using oscillators, which generate a repetitive electrical signal with steady timing.

In classical systems, the state of an oscillator is represented by the